

Empirical Methods in Finance

Has the Structure of Risk in Financial Markets Changed Since COVID-19?

EMiF Project Report

Group members: *CHRAIBI Saad*
CHEKARA Ishaq
ABAROUDI Yazid
JEBLI Omar

Course: Empirical Methods in Finance

Instructor: Florian Ielpo

Institution: HEC Lausanne

Submission date: *31 May 2026*

This report uses only the data contained in `Data.xlsx`. All empirical results are reproducible from the accompanying Jupyter notebook.

Abstract

This report studies whether the structure of risk in financial markets changed after the COVID-19 shock. We define risk structure as the joint organization of risk levels, tail risk, cross-asset dependence, common risk factors, transmission channels, regimes and portfolio diversification. Using the daily data in `Data.xlsx`, we compare a pre-COVID sample ending on 10 March 2020 with a post-COVID sample starting on 11 March 2020. The evidence supports a partial change in risk structure. Risk did not rise uniformly across all assets, and average cross-asset correlation even declined slightly from 0.102 to 0.084. However, the composition of dependence changed materially: equity-credit correlation increased, equity-rate correlation decreased, credit internal correlation strengthened, and the first two principal components rose from 43.45% to 50.32% of total variance. Portfolio evidence points in the same direction: the pre-COVID minimum-variance allocation became riskier post-COVID and its diversification ratio fell from 2.94 to 2.33. Robustness checks based on alternative COVID breakpoints, moving-block bootstrap, EWMA dynamic covariance, Student-t GARCH and corrected Granger tests support the same conclusion. We therefore answer the research question as follows: yes, but partially. The post-COVID change is better described as a reorganization of market risk than as a simple increase in all forms of risk.

Keywords: COVID-19, risk structure, volatility, correlation, PCA, GARCH, EWMA, Granger causality, diversification, ERC.

Contents

1	Introduction	3
2	Literature and hypotheses	3
3	Data and transformations	4
3.1	Dataset	4
3.2	Return definitions	5
4	Methodology	5
4.1	Risk levels and tail risk	5
4.2	Correlation and covariance structure	6
4.3	Principal component analysis	6
4.4	Regimes, GARCH and dynamic covariance	6
4.5	Transmission and portfolio implications	7
5	Empirical results	7
5.1	Risk levels changed selectively, not uniformly	7
5.2	The average correlation fell, but the correlation structure changed	8
5.3	PCA shows higher common-factor concentration	9
5.4	Rolling evidence points to unstable regimes	10
5.5	Portfolio diversification deteriorated for some allocations	12
6	Robustness checks	13
6.1	Alternative COVID breakpoints	13
6.2	GARCH and Student-t density check	13
6.3	Granger transmission with multiple-testing correction	13
6.4	No-oil, no-yields and weekly robustness	14
7	Discussion	14
8	Conclusion	14

1. Introduction

The COVID-19 shock was not only a large market drawdown. It was also a test of the way risks interact across equities, bonds, credit, commodities and foreign exchange. A simple before-after comparison of prices would therefore be incomplete. The relevant question for this project is whether the *structure* of risk changed: did volatility and tail risk change, did cross-asset correlations move, did common factors become more concentrated, and did portfolio diversification behave differently?

We answer this question empirically using the dataset provided in `Data.xlsx`. The empirical design follows the logic of the EMiF course. First, the analysis starts from return transformations and stylized facts of financial returns: volatility is time-varying, distributions are non-Gaussian, and squared or absolute returns often display persistence (Cont, 2001). Second, it studies co-movement using covariance and correlation matrices. Third, it uses PCA to measure whether a smaller number of common shocks explains more of the cross-section. Fourth, it studies regimes, dynamic covariance and risk transmission using rolling estimates, EWMA covariance, GARCH and Granger-type tests. Finally, it translates the results into portfolio terms through minimum-variance and equal-risk-contribution allocations.

The report reaches a deliberately nuanced conclusion. The evidence does not support the statement that risk increased everywhere after COVID-19. Some volatility measures increased, others fell. The average correlation across all assets declined slightly. But this average masks a more important reallocation of risk: equity-credit links strengthened, equity-rate links weakened, credit risk became more internally synchronized, and PCA concentration rose. The post-COVID environment therefore looks less like a uniform risk increase and more like a selective reorganization of risk channels.

2. Literature and hypotheses

The literature motivates four testable ideas. First, financial returns are not well described by a constant-volatility Gaussian model. They display heavy tails, volatility clustering, leverage effects and changing dependence (Cont, 2001). A COVID shock can therefore affect the conditional distribution of returns, not just their mean.

Second, crises often change diversification properties. Correlations can increase in stressed markets, and diversification can fail when it is most needed (Longin and Solnik, 2001). This does not mean every correlation must rise; rather, the relevant object is the structure of the covariance matrix. A weaker average correlation can still coexist with stronger links in key blocks, such as equities and credit.

Third, the EMiF course emphasizes that volatility is persistent and should be modeled conditionally. ARCH and GARCH models were introduced by Engle (1982) and Bollerslev (1986); Student-t GARCH is especially relevant when residuals remain heavy-tailed (Bollerslev, 1987). We therefore treat GARCH as a robustness check for conditional volatility dynamics, not as the main forecasting exercise.

Fourth, risk regimes and breaks matter. Structural-break methods such as Bai and Perron (1998) provide a formal way to think about parameter instability, while Markowitz-type portfolios are highly sensitive to changes in covariance estimates (Markowitz, 1952). Equal-risk-contribution

portfolios add a portfolio-risk perspective by focusing on risk contributions rather than capital weights (Maillard et al., 2010). We use these ideas to evaluate whether the same allocation became less diversified after COVID-19.

These ideas lead to five hypotheses:

1. **Risk levels:** post-COVID volatility and downside risk should change for some assets, especially equities, rates, credit and commodities.
2. **Co-movement:** the correlation matrix should change even if the average correlation does not rise uniformly.
3. **Common factors:** PCA should show whether risk became more concentrated in fewer components.
4. **Regimes and transmission:** rolling estimates, EWMA covariance and Granger tests should reveal whether risk dynamics became more unstable or more connected.
5. **Portfolio implications:** if the risk structure changed, pre-COVID allocations should produce different post-COVID risk and diversification properties.

3. Data and transformations

3.1. Dataset

The raw dataset contains 9,601 daily observations and 14 financial series from 31 December 1989 to 24 April 2026. The assets cover major equity indices, government yields, commodities, foreign exchange and credit indices. The analysis uses a balanced panel after transformations with 7,332 observations from 4 January 1990 to 24 April 2026.

The COVID breakpoint is 11 March 2020, corresponding to the World Health Organization pandemic announcement and the transition into the major market stress period. The pre-COVID sample contains 6,064 observations from 4 January 1990 to 10 March 2020. The post-COVID sample contains 1,268 observations from 11 March 2020 to 24 April 2026.

Table 1: Data treatment summary

Item	Treatment
Frequency	Daily observations from <code>Data.xlsx</code>
Main sample	4 January 1990 to 24 April 2026 after transformations
COVID breakpoint	11 March 2020
Price/index/FX/bond-price series	Log returns
Government yields	Daily changes in basis points
Oil futures	Asinh difference, to handle the April 2020 negative WTI episode
Missing values	Balanced panel after transformation, using drop of rows with missing values
Reason for balanced panel	Comparable covariance, correlation and PCA estimates require aligned observations

3.2. Return definitions

For most price and index series, we use log returns:

$$r_{i,t} = \log(P_{i,t}) - \log(P_{i,t-1}). \quad (1)$$

For government yields, the relevant variable is a daily change in basis points:

$$\Delta y_{i,t} = 100 \times (y_{i,t} - y_{i,t-1}). \quad (2)$$

Oil futures require special treatment because WTI briefly traded at negative prices in April 2020. Since logarithms are not defined for negative prices, we use an inverse-hyperbolic-sine difference:

$$r_t^{oil} = \text{asinh}(P_t) - \text{asinh}(P_{t-1}). \quad (3)$$

This keeps the oil series in the analysis while avoiding an artificial deletion of a central COVID-period event. It also means that oil volatility is interpreted cautiously in the robustness discussion.

4. Methodology

4.1. Risk levels and tail risk

Annualized volatility is computed as

$$\sigma_{ann,i} = \sqrt{252} \sigma_i, \quad (4)$$

for return-like series. We also compute skewness, excess kurtosis, 5% value-at-risk and 5% conditional value-at-risk. CVaR is used because it focuses on the average loss beyond the VaR threshold and is therefore more informative about tail losses.

Differences between pre- and post-COVID risk levels are assessed with bootstrap confidence intervals. Because daily returns are not iid in the strict sense, the final notebook also includes moving-block bootstrap robustness for PCA concentration.

4.2. Correlation and covariance structure

Cross-asset dependence is measured through the correlation matrix

$$\rho_{ij} = \frac{\text{Cov}(r_i, r_j)}{\sqrt{\text{Var}(r_i) \text{Var}(r_j)}}. \quad (5)$$

We compare individual pairs, block correlations and the global matrix difference. The global difference is summarized using the Frobenius norm:

$$\|R_{post} - R_{pre}\|_F = \sqrt{\sum_{i,j} (\rho_{ij}^{post} - \rho_{ij}^{pre})^2}. \quad (6)$$

This measure is useful because a small average-correlation change can still hide large changes in specific areas of the matrix.

4.3. Principal component analysis

PCA is applied to the standardized return panel. Let R be the correlation matrix and let λ_k be its eigenvalues. The share of variance explained by component k is

$$s_k = \frac{\lambda_k}{\sum_j \lambda_j}. \quad (7)$$

The first component captures the dominant common risk factor. The cumulative share of PC1 and PC2 measures the degree of risk concentration in the first two dimensions.

4.4. Regimes, GARCH and dynamic covariance

Rolling volatility and rolling average correlations are used to visualize time-varying risk. For conditional volatility robustness, we estimate a GARCH(1,1) model,

$$r_t = \mu + \epsilon_t, \quad (8)$$

$$\epsilon_t = \sigma_t z_t, \quad (9)$$

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (10)$$

Persistence is summarized by $\alpha + \beta$. A Student-t GARCH density check is included for the S&P500 to account for heavy tails.

Dynamic covariance is also studied through an EWMA estimator:

$$\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) r_t r_t'. \quad (11)$$

This is not a full DCC model, but it provides a course-consistent way to check whether the covariance structure evolves over time.

4.5. Transmission and portfolio implications

Pairwise Granger tests are used as exploratory evidence on lead-lag transmission. To avoid overclaiming, the notebook reports raw p-values and multiple-testing corrections using FDR and Bonferroni logic.

Portfolio implications are evaluated through equal-weight, minimum-variance and equal-risk-contribution allocations. If w is a vector of weights and Σ is the covariance matrix, portfolio volatility is

$$\sigma_p = \sqrt{w' \Sigma w}. \quad (12)$$

The diversification ratio is

$$DR = \frac{\sum_i w_i \sigma_i}{\sqrt{w' \Sigma w}}. \quad (13)$$

A decline in this ratio suggests weaker diversification for the same allocation.

5. Empirical results

5.1. Risk levels changed selectively, not uniformly

Table 2 summarizes the main bootstrap results for volatility. S&P500 volatility increased from 17.35% to 20.67%, a significant rise of 3.32 percentage points. German 10-year yield volatility also increased significantly, from 69.28 to 81.93 basis points annualized. Gold, US investment-grade credit and US high-yield credit also display significant volatility increases. By contrast, SMI, EURUSD, USDJPY and USDCHF show significant volatility decreases.

Table 2: Selected volatility-change tests

Asset	Pre	Post	Change	p-value	Interpretation
S&P500	0.1735	0.2067	0.0332	0.0067	increased
Eurostoxx 50	0.2077	0.1989	-0.0088	0.3533	not significant
SMI	0.1743	0.1496	-0.0247	0.0067	decreased
US 10Y yield	92.5748	96.2001	3.6253	0.2467	not significant
German 10Y yield	69.2791	81.9287	12.6496	0.0000	increased
Oil futures	0.3771	4.8371	4.4600	0.0000	increased
Gold	0.1557	0.1793	0.0236	0.0000	increased
US IG Bonds	0.0468	0.0653	0.0185	0.0000	increased
US HY Bonds	0.0409	0.0627	0.0217	0.0000	increased
USDCHF	0.1137	0.0806	-0.0330	0.0000	decreased

The result is already more subtle than a simple “risk increased” story. Some risky assets saw higher volatility, but some equity and FX series saw lower volatility in the post-COVID average. Tail risk gives a similar message: German yields, oil, credit and some FX series changed significantly, while the S&P500 CVaR increase is not statistically significant.

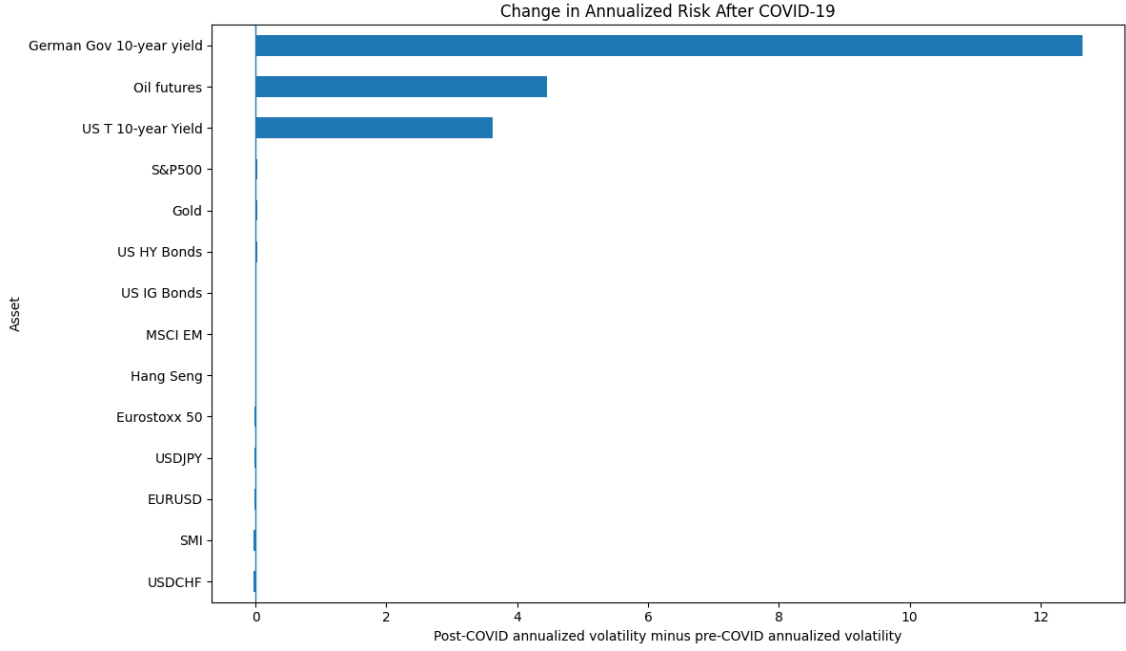


Figure 1: Change in annualized volatility between the pre-COVID and post-COVID samples.

5.2. The average correlation fell, but the correlation structure changed

Average pairwise correlation declined from 0.1022 to 0.0842. On its own, this would suggest no general correlation breakdown. But the block analysis in Table 3 shows that this average hides important changes. Equity-credit correlation increased from 0.1375 to 0.3142, while equity-rate correlation fell from 0.1620 to 0.0346. Credit within-correlation rose sharply from 0.2503 to 0.5566, and rates-credit correlation became more negative.

Table 3: Block-correlation changes

Block pair	Pre	Post	Change	p-value	Interpretation
Equity-equity	0.4783	0.4550	-0.0233	0.4933	not significant
Equity-rates	0.1620	0.0346	-0.1274	0.0000	decreased
Equity-credit	0.1375	0.3142	0.1767	0.0000	increased
Equity-commodity	0.0641	0.0440	-0.0201	0.5733	not significant
Rates-credit	-0.3339	-0.4585	-0.1246	0.0000	decreased

The pairwise tests confirm this pattern. The S&P500 vs US IG Bonds correlation increased from -0.1412 to 0.1905. The S&P500 vs US HY Bonds correlation increased from 0.2633 to 0.5878. US IG vs US HY correlation increased from 0.2503 to 0.5566. These changes are statistically significant under Fisher correlation tests.

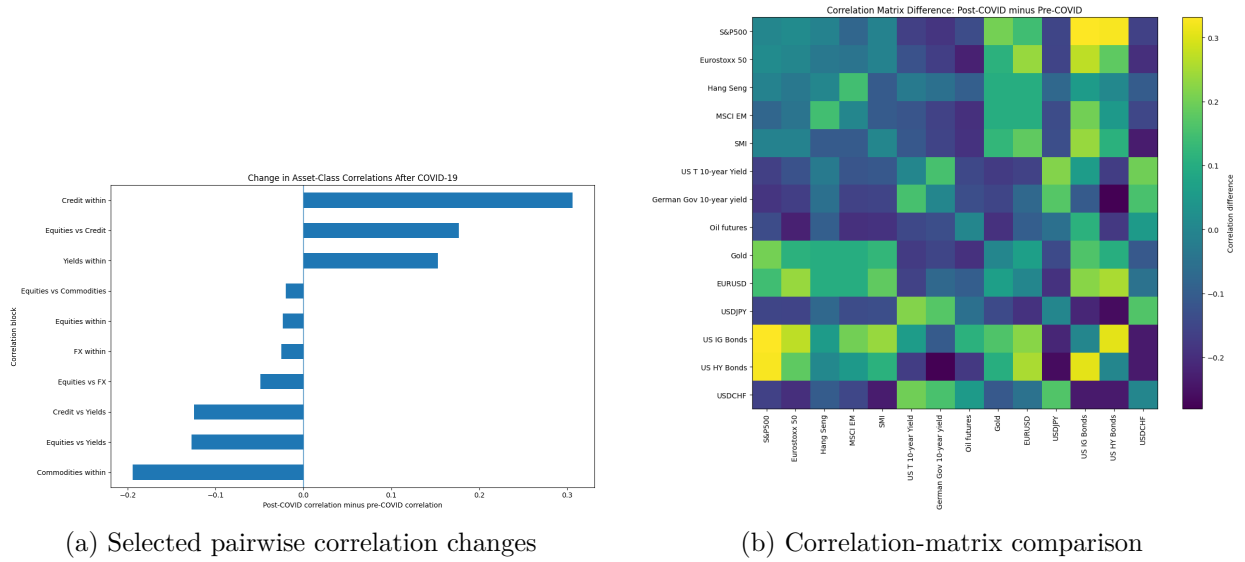


Figure 2: Changes in cross-asset dependence.

The global Frobenius norm of the correlation-matrix difference is 2.179, with bootstrap interval $[1.936, 2.622]$. This supports the view that the matrix changed materially even though the average correlation did not rise.

5.3. PCA shows higher common-factor concentration

The PCA results are one of the clearest pieces of evidence for a structural change. PC1 explained 25.81% of variance before COVID and 29.46% after COVID. PC1 plus PC2 increased from 43.45% to 50.32%. Bootstrap inference classifies both increases as statistically significant.

Table 4: PCA explained variance comparison

Component	Pre	Post	Change
PC1	0.2581	0.2946	0.0365
PC2	0.1764	0.2086	0.0322
PC3	0.1387	0.1123	-0.0264
PC4	0.0728	0.0863	0.0135
PC5	0.0682	0.0720	0.0038
PC1 + PC2	0.4345	0.5032	0.0687

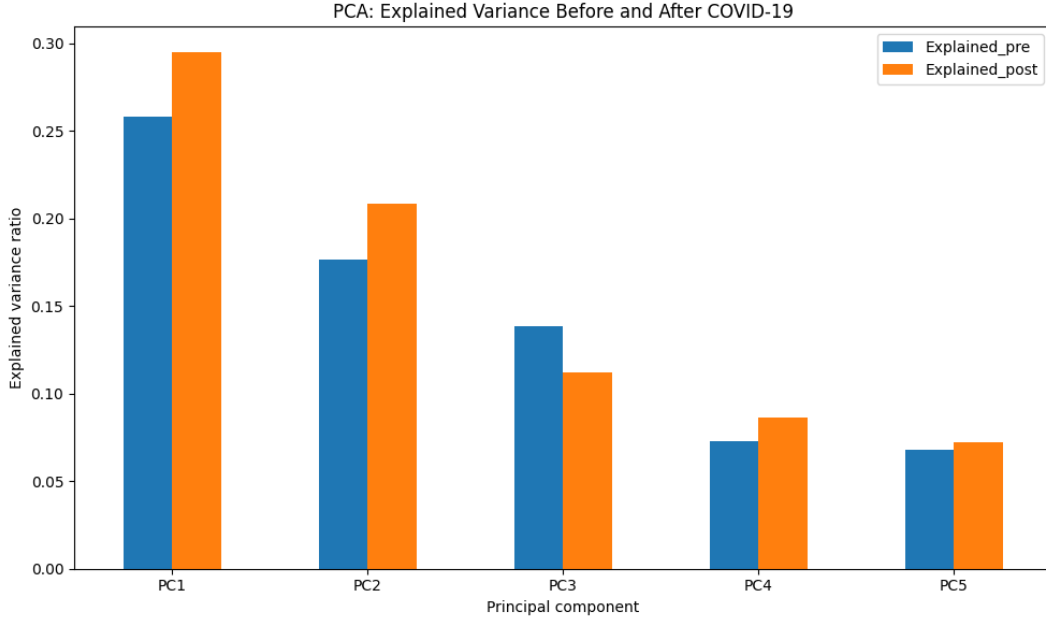


Figure 3: Pre- and post-COVID PCA explained variance.

Economically, this means that more of the cross-section became explained by the first two common dimensions. This is a meaningful change for risk monitoring: a portfolio manager cannot only track standalone volatility; common-factor exposure became more important after COVID.

5.4. Rolling evidence points to unstable regimes

Rolling volatility and rolling average correlations show that the COVID period was not a one-day outlier. Risk moved through regimes, with volatility spikes followed by normalization at different speeds across asset classes. The rolling average correlation does not remain permanently high, which helps explain why the full-sample post-COVID average correlation declined. The issue is therefore not a permanent rise in all correlations, but a change in when and where correlations matter.

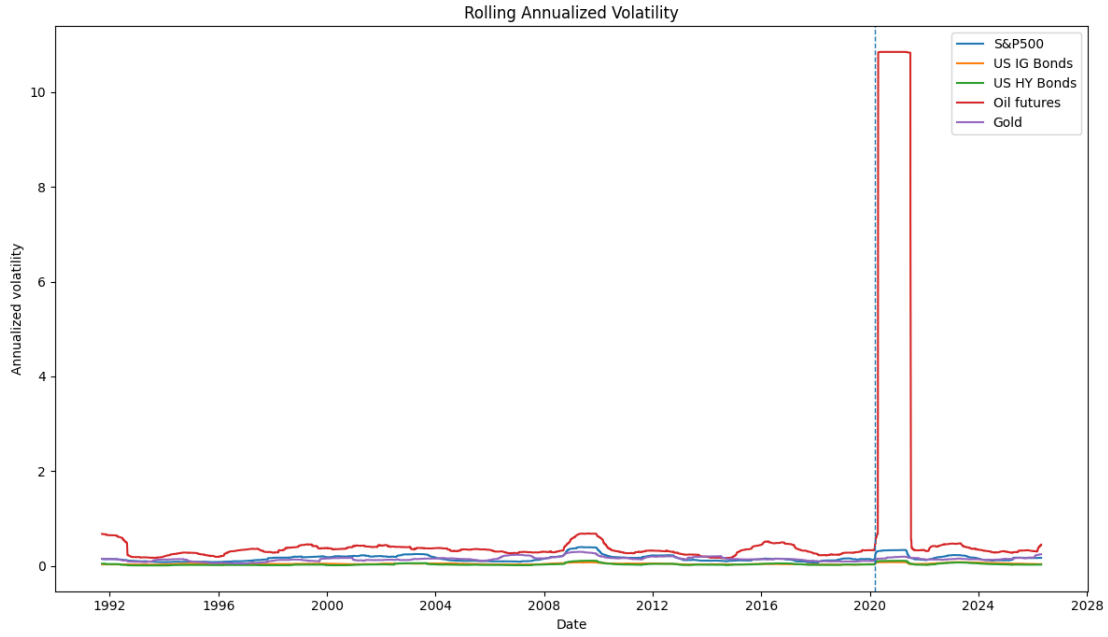


Figure 4: Rolling annualized volatility.

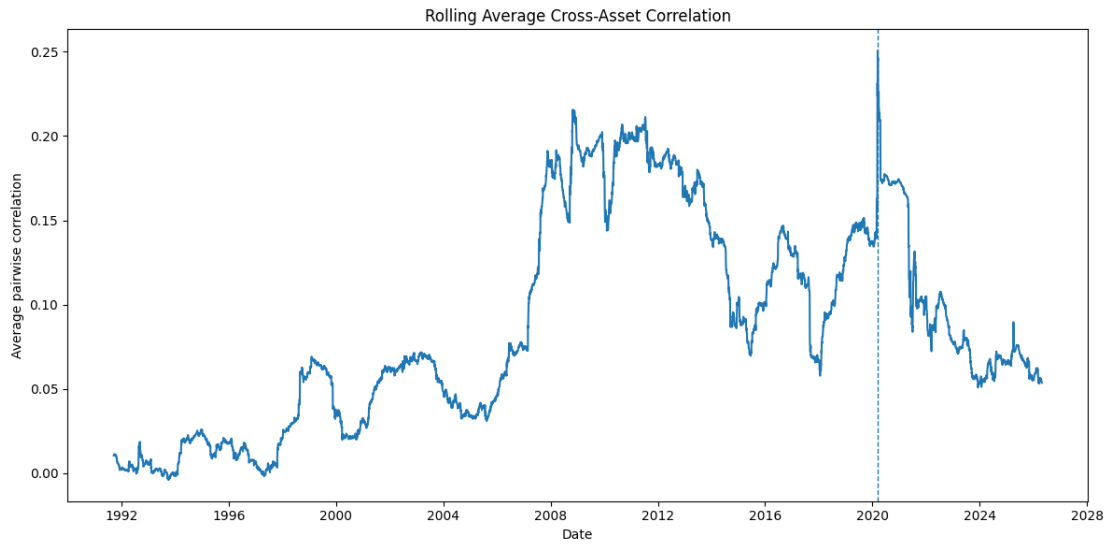


Figure 5: Rolling average cross-asset correlation.

The equity-bond correlation regime is particularly important. The S&P500 vs US IG Bonds relationship moved from negative to positive, which is a direct challenge to a traditional diversification argument. Figure 6 shows that the rolling equity-credit correlation changed materially around the post-COVID period.

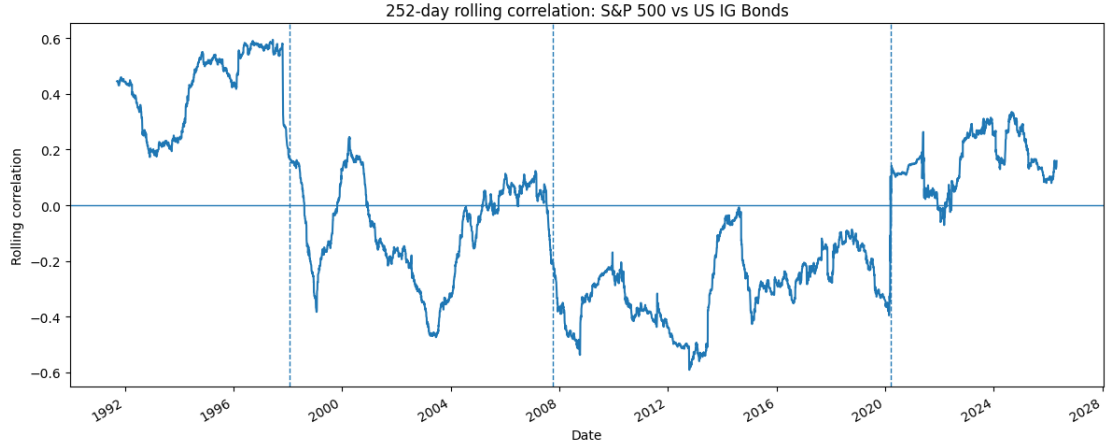


Figure 6: Rolling S&P500 vs US IG Bonds correlation and detected regimes.

5.5. Portfolio diversification deteriorated for some allocations

Portfolio evidence translates the statistical results into investment terms. The pre-COVID minimum-variance portfolio had a pre-COVID volatility of 2.51%, but the same weights produced 3.17% volatility post-COVID. Its diversification ratio fell from 2.94 to 2.33. The equal-risk-contribution portfolio was more stable, with volatility moving from 4.35% to 4.48%, but its diversification ratio also declined from 2.60 to 2.40.

Table 5: Portfolio risk using pre-COVID weights in the post-COVID covariance matrix

Portfolio	Pre vol	Post vol	Vol change	DR pre to post
Equal weight	0.0744	0.0762	0.0019	1.8809 to 1.8204
Minimum variance	0.0251	0.0317	0.0067	2.9427 to 2.3264
Equal risk contribution	0.0435	0.0448	0.0013	2.6021 to 2.3980

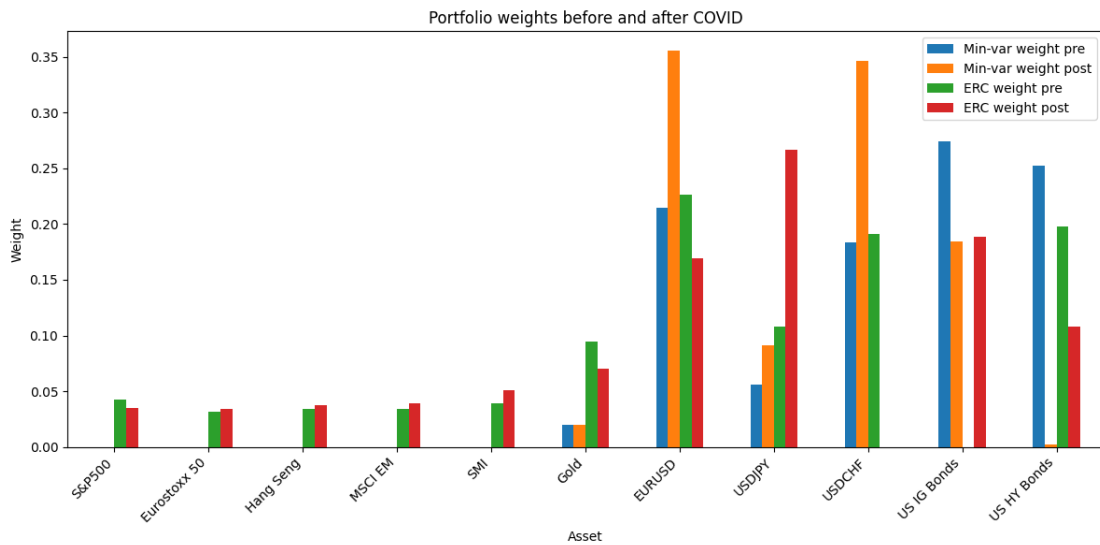


Figure 7: Pre- and post-COVID portfolio weights.

The portfolio interpretation is central to the research question. A change in risk structure

should affect diversification. This is exactly what the results show: even when average correlations do not rise, the covariance matrix changes enough to alter the risk of allocations calibrated before COVID.

6. Robustness checks

6.1. Alternative COVID breakpoints

The conclusions are not driven by a single chosen date. Table 6 compares four plausible breakpoints: 24 February 2020, 11 March 2020, 23 March 2020 and 1 June 2020. Across these alternatives, PC1 increases by roughly 3.3 to 3.7 percentage points and PC1+PC2 increases by roughly 5.4 to 7.8 percentage points. This confirms that the PCA concentration result is not an artifact of one date.

Table 6: Breakpoint robustness

Breakpoint	Avg corr pre	Avg corr post	Corr change	PC1 pre	PC1 post	PC1+PC2 change
2020-02-24	0.0986	0.0938	-0.0048	0.2550	0.2877	0.0778
2020-03-11	0.1022	0.0842	-0.0180	0.2581	0.2946	0.0687
2020-03-23	0.1056	0.0736	-0.0320	0.2579	0.2913	0.0583
2020-06-01	0.0934	0.0760	-0.0173	0.2547	0.2921	0.0544

6.2. GARCH and Student-t density check

The GARCH(1,1) robustness check confirms that conditional volatility changed for representative assets. Average conditional volatility increased for the S&P500, US IG Bonds, US HY Bonds and oil. For the S&P500, average GARCH conditional volatility rose from 15.51% to 17.57%. For credit, the increase is also visible: US IG moved from 4.55% to 6.01%, and US HY from 3.73% to 4.91%.

Table 7: GARCH(1,1) robustness summary

Asset	α pre	α post	β pre	β post	Avg cond. vol change
S&P500	0.1194	0.1588	0.8636	0.8040	2.0557
US IG Bonds	0.0439	0.0968	0.9470	0.8816	1.4628
US HY Bonds	0.0615	0.1727	0.8891	0.8263	1.1809
Oil futures	0.0994	0.0035	0.8945	0.9758	71.4832

These estimates are interpreted descriptively. Because the project is not primarily a volatility-forecasting exercise, the GARCH results are not presented as a full out-of-sample model comparison. Their role is to check whether the volatility conclusions remain visible in a conditional-volatility framework. The Student-t density check strengthens the link with the course by acknowledging that Gaussian GARCH can understate heavy-tailed residual behavior.

6.3. Granger transmission with multiple-testing correction

The Granger section studies risk transmission among S&P500, US IG Bonds, US HY Bonds, Gold and Oil futures. Several post-COVID links appear in raw tests, for example US IG Bonds to

S&P500 and oil to S&P500. However, because many pairwise tests are performed, the corrected results are treated as exploratory evidence rather than proof of economic causality.

The main lesson is not that the report has discovered true causal relations. The safer interpretation is that lead-lag patterns became more visible in some stress-sensitive pairs after COVID, especially around credit and commodities. The use of FDR and Bonferroni corrections prevents the analysis from overstating weak p-values.

6.4. No-oil, no-yields and weekly robustness

Oil is a special case because of the negative WTI episode in 2020. Excluding oil leaves the main conclusion unchanged: the average correlation change is almost zero, but PC1 still increases from 27.40% to 31.72%, and PC1+PC2 also rises. Excluding yield-change series also leaves PCA concentration higher post-COVID. At weekly frequency, PC1 rises even more, from 29.49% to 40.49%, although weekly aggregation changes the interpretation of tail risk and volatility.

These robustness checks are important because they show that the central result is not an artifact of oil, yields or daily frequency. The post-COVID change is mainly a change in dependence and factor concentration.

7. Discussion

The empirical evidence points to a partial transformation of market risk. Three findings are especially important.

First, standalone volatility does not deliver a single story. Some assets became riskier, especially S&P500, German yields, gold and credit. Other assets became less volatile on average, including SMI and several FX series. This is why the conclusion should not be phrased as “risk increased everywhere.”

Second, correlation averages are misleading. The overall average correlation declined slightly, but this hides large block-level changes. Equity-credit co-movement increased, equity-rate co-movement weakened, and credit became more internally correlated. For a risk manager, this is more relevant than the simple average correlation.

Third, common-factor concentration increased. PC1 and PC1+PC2 explained more variance after COVID. This is a classic sign that risk became more concentrated in broad common shocks. For portfolio management, this means that nominal diversification across many assets may overstate true diversification if the assets load on the same dominant factors.

The portfolio results make the same point in practical terms. The pre-COVID minimum-variance portfolio became riskier when evaluated with the post-COVID covariance matrix, and its diversification ratio fell sharply. Equal-risk-contribution allocations were more robust, but not immune. This matters because the project question is ultimately about risk structure, not only statistical significance.

8. Conclusion

The answer to the research question is: **yes, but partially**. The post-COVID period did not produce a uniform rise in every measure of risk. It produced a selective reorganization of risk.

Volatility increased significantly for several assets, especially S&P500, German yields, gold and credit, but not for all markets. The average correlation fell from 0.1022 to 0.0842, but this masks important reallocations in the correlation matrix. Equity-credit correlation increased, equity-rate correlation declined, and credit internal correlation strengthened. PCA shows that common-factor concentration increased: PC1 rose from 25.81% to 29.46%, and PC1+PC2 rose from 43.45% to 50.32%.

The portfolio evidence confirms the economic relevance of these changes. A pre-COVID minimum-variance allocation became riskier post-COVID and its diversification ratio fell from 2.94 to 2.33. This means the change is not only statistical; it affects portfolio construction and risk monitoring.

The main limitation is that structural breaks are studied through breakpoint robustness and rolling evidence rather than through a full formal Bai-Perron or Markov-switching model. The GARCH analysis is also descriptive rather than a complete volatility-forecasting exercise. These limits do not overturn the conclusion. They mainly define the scope of the evidence: the project supports a partial change in risk structure, especially through dependence, factor concentration and diversification channels.

References

- Bai, J., and Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1), 47–78.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *Review of Economics and Statistics*, 69(3), 542–547.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Engle, R. F. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Engle, R. F. (2002). Dynamic conditional correlation: a simple class of multivariate GARCH models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3), 424–438.
- Longin, F., and Solnik, B. (2001). Extreme correlation of international equity markets. *Journal of Finance*, 56(2), 649–676.
- Maillard, S., Roncalli, T., and Teiletche, J. (2010). The properties of equally weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4), 60–70.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- J.P. Morgan/Reuters (1996). *RiskMetrics Technical Document*. Fourth edition.